

AI Hydra — Project Roadmap & Artifact Registry

Version: 1.1 — June 2026. Gen 1/2/3 doctrine formalised. **Owner:** Darren Humphries, Group CISO **Purpose:** Master reference linking all project artifacts, their relationships, citation chains, and development status. Start here to understand what exists, where it lives, and what depends on what.

0. Framework Doctrine

The AI Hydra framework is an offensive capability proliferation tracker — the cyber equivalent of nuclear proliferation monitoring. It tracks whether threat actors have weaponised specific capabilities, not whether the underlying architecture exists in benign form.

Gen 1: Component Proliferation — "Can they build each part?" 34 individual offensive capabilities tracked. Like enriching uranium, building centrifuges, manufacturing missile bodies. Each component tracked for offensive demonstration. Enterprise IT doing the same thing (AIOps self-healing, ITSM automation) is dual-use context, not evidence.

Gen 2: System Integration — "Does it all work together?" 8 orchestration capabilities that emerge when Gen 1 components combine. Like putting the warhead on the missile. Evidence requires orchestrated combination of multiple Gen 1 components in an offensive system.

Gen 3: Operational Deployment — "Is this now standard arsenal?" (Future) Commodity tradecraft for Tier 1-2 actors. Like having a reliable nuclear arsenal with doctrine and trained operators. The 24-month horizon.

Evidence threshold: Only offensive demonstrations change gap status. The tracker watches the weapons programme, not the energy programme.

1. Project Timeline

2026-02-04	Think tank working group established
2026-03-27	Framework created (Layer 1: 34 capabilities)
2026-04-15	Mythos Preview / Glasswing findings integrated
2026-05-01	Gap analysis completed (17/9/8)
2026-05-11	GTIG AI Threat Tracker → 3 gaps close (17/12/5)

2026-05-11 Mini Shai-Hulud Wave 5 case study added
 2026-05-14 Shai-Hulud LinkedIn post published
 2026-06-03 CleverHans adaptive worm research published (arXiv 2606.03811)
 2026-06-04 Presented at Infosec Europe
 2026-06-04 Framework + gap analysis LinkedIn post published
 2026-06-04 Full keynote deck LinkedIn post published
 2026-06-04 CleverHans gap analysis completed → 2 more gaps close
 2026-06-04 Portfolio Strategy (chess model) developed
 2026-06-04 Gen 2 Capabilities Framework defined
 2026-06-06 GitHub tracker app built with 25 intel cards
 2026-06-06 Skills created (Framework Skill, Research Skill)

2. Artifact Registry

2.1 Core Framework Documents

Artifact	Location	Purpose	Status
Speech Research Brief	Project: <code>Speech_Research_Brief_AI_Hydra.md</code> / <code>.docs</code>	Full keynote research with data points, speaker notes, Q&A prep	Complete
CISO Voice Reference	Project: <code>CISO_Voice_Reference.md</code>	Editorial voice, analytical patterns, LinkedIn structure	Complete
Vulnerability Analysis Skill	Project: <code>Vulnerability_Analysis_Skill.md</code>	Methodology for CVE/vuln analysis and publication	Complete
AI Hydra Framework Skill	Project: <code>AI_Hydra_Framework_Skill.md</code>	Framework update methodology, 34 capabilities, 25 intel entries, current state baseline	Complete
AI Hydra Research Skill	Project: <code>AI_Hydra_Research_Skill.md</code>	Proactive research methodology, source hierarchy, search strategies, sweep cadence	Complete

2.2 Evolved Framework Documents

Artifact	Location	Purpose	Status
Hydra Portfolio Strategy	Project: <code>Hydra_Portfolio_Strategy.md</code>	Chess model — multi-stage, multi-campaign, Pawns/Knights/King risk tiers	Complete
Gen 2 Capabilities Framework	Project: <code>Gen2_Capabilities_Framework.md</code>	8 Gen 2 capabilities that emerge from	Complete

Artifact	Location	Purpose	Status
		Layer 1 maturity, probabilistic Sankey model	
CleverHans Gap Analysis	Project: <code>CleverHans_vs_Hydra_Gap_Analysis.md</code>	Full mapping of CleverHans research against Layer 1 and Gen 2	Complete
CleverHans Research Update	Project: <code>CleverHans_Research_Update_Gap_Changes.md</code>	Citations, gap changes, intel card, JSON updates ready to commit	Complete

2.3 Presentation Assets

Artifact	Location	Purpose	Status
Keynote Deck (19 slides)	Output: <code>The_Convergence_Keynote.pptx</code>	Infosec Europe presentation — opening questions, three lanes, architecture, gap analysis, Shai-Hulud, tier mapping, closing	Complete
Architecture + Gap Slides	Output: <code>Hydra_Architecture_Slides_Updated.pptx</code>	Two standalone slides with GTIG-updated gap analysis (17/12/5)	Complete
Architecture Graphic (React)	Output: <code>AI_Hydra_Tracker.jsx</code>	Interactive preview of the tracker app	Complete

2.4 GitHub Tracker

Artifact	Location	Purpose	Status
Tracker App	Output: <code>ai-hydra-tracker.zip</code> → GitHub repo	GitHub Pages site — architecture, gap analysis, 25 intel cards, timeline	Ready to push
gaps.json	<code>ai-hydra-tracker/data/gaps.json</code>	34 Layer 1 capabilities with status	Needs CleverHans update
intel.json	<code>ai-hydra-tracker/data/intel.json</code>	25 intel cards with analysis + capability impact	Needs CleverHans card
changelog.json	<code>ai-hydra-tracker/data/changelog.json</code>	8 timeline entries	

Artifact	Location	Purpose	Status
			Needs CleverHans entry

2.5 LinkedIn Posts

Post	Topic	Status
Mini Shai-Hulud supply chain worm	Self-propagating supply chain worm, five-wave evolution, "it doesn't think yet"	Published
CVE noise / vulnerability prioritisation	36→4 reduction, decision tree, CVSS vs reality	Published
Framework + gap analysis (Infosec Europe)	Architecture and gap slides attached, framework summary	Published
Full keynote deck (Infosec Europe)	Complete 19-slide deck with two paragraphs	Published
CleverHans "the thesis is no longer theoretical"	Not yet written	Pending

2.6 Supporting Research

Document	Location	Purpose
CVE Noise Analysis	Upload: <code>Why_Raw_CVE_Counts_Are_Noise.docx</code>	Production audit — 36→4 reduction, decision tree, CVSS vs reality
Iran Intel Briefing	Project: <code>Intel_Briefing.pdf</code>	Iran threat landscape, tier model, APT group mapping
Patch Tuesday Brief	Project: <code>MS_Patch_Tuesday_May2026_Brief_v4.docx</code>	Microsoft vulnerability analysis methodology
LinkedIn Defender Post	Project: <code>LinkedIn_Defender_Pattern_Post.md</code>	Defender pattern analysis post

3. Citation Chain — What References What

3.1 The Thesis Chain

CISO Voice Reference

- ↳ Speech Research Brief (voice + analytical patterns applied)
 - ↳ Keynote Deck (visual presentation of the brief)
 - ↳ LinkedIn Posts (distilled for platform)
 - ↳ GitHub Tracker (living version of the framework)
 - ↳ Framework Skill (methodology to update tracker)
 - ↳ Research Skill (methodology to find new intel)

3.2 The Framework Evolution Chain

Layer 1: Original 34 Capabilities (March 2026)

↳ Gap Analysis (May 2026): 17/9/8

↳ GTIG Update (11 May 2026): 17/12/5 – 3 gaps closed

↳ CleverHans Update (4 June 2026): 2 more gaps closed

↳ Portfolio Strategy: reframes from linear to chess model

↳ Gen 2 Framework: 8 capabilities emerge from Layer 1

3.3 The Intelligence Citation Map

Each intel entry and what it evidences:

INITIAL ACCESS & RECON

- └─ ia-1 (Autonomous vuln scanning): EXISTS
 - └─ intel-xbow (XBOW #1 HackerOne)
 - └─ intel-h3-growth (Horizon3 180K+ pentests)
 - └─ intel-pentagi (PentAGI open source)
 - └─ intel-darpa-aixcc (DARPA AIXCC \$152/task)
 - └─ intel-artemis (ARTEMIS beats 9/10 humans)
- └─ ia-2 (Autonomous exploit execution): EXISTS
 - └─ intel-xbow
 - └─ intel-h3-growth
 - └─ intel-h3-cloud (NodeZero Global Admin without CVEs)
 - └─ intel-pentagi
 - └─ intel-artemis
 - └─ intel-gtig-gtg1002 (GTG-1002 80% autonomous)
- └─ ia-4 (Zero-day discovery): EXISTS
 - └─ intel-mythos (Mythos every major OS)
 - └─ intel-openai-high (GPT 5.3-Codex "High" threshold)
 - └─ intel-frontier-trajectory (0→73% in 12 months)
- └─ ia-5 (Zero-day weaponisation, logic flaws): PARTIAL
 - └─ intel-gtig-zero-day (first AI zero-day)
 - └─ intel-openai-high
 - └─ intel-frontier-trajectory

CREDENTIAL & LATERAL MOVEMENT

- └─ cl-3 (AI-driven lateral movement): PARTIAL
 - └─ intel-darpa-aixcc (ATLANTIS)
 - └─ intel-h3-hvt (NodeZero HVT)
 - └─ intel-h3-cloud (cross-domain pivoting)
 - └─ intel-artemis
 - └─ intel-cleverhans-worm* (pending)

AI ORCHESTRATION

- └─ ao-1 (Attack surface assessment / Recursive reasoning): EXISTS→PARTIAL*
 - └─ intel-xbow, intel-h3-growth, intel-pentagi
 - └─ intel-cleverhans-worm* (recursive reasoning demonstrated)
- └─ ao-2 (Real-time tool selection): PARTIAL
 - └─ intel-pentagi (multi-agent)
 - └─ intel-promptsby (GeminiAutomationAgent)
 - └─ intel-h3-hvt (NodeZero HVT)
 - └─ intel-china-claude (80-90% autonomous ops)
 - └─ intel-cleverhans-worm* (reinforcing)

- └─ ao-3 (Environment value assessment): GAP
 - └─ (no evidence – hardest gap)
- └─ ao-5 (Adaptive evasion / Field-to-foundry): GAP→PARTIAL*
 - └─ intel-promptsy (dynamic C2, evasion overlays)
 - └─ intel-canfail (AI decoy code)
 - └─ intel-cleverhans-worm* (CVE integration within hours)
- └─ ip-1 (Selective IP targeting): GAP
 - └─ (no evidence)

HYBRID C2

- └─ c2-1 (C2 via legitimate cloud): EXISTS
 - └─ intel-apt34-c2 (OneDrive/Exchange)
 - └─ intel-apt35-c2 (Dropbox/GDrive/IPFS)
 - └─ intel-muddywater-c2 (legitimate admin tools)
 - └─ intel-shai-hulud (GitHub API, Session Protocol)
- └─ c2-4 (Dynamic C2 rotation): PARTIAL
 - └─ intel-promptsy (runtime infrastructure updates)
- └─ c2-5 (Legitimate user traffic mimicry): GAP
 - └─ intel-lumu-aic2 (prediction only)

EVASION

- └─ ev-1 (Polymorphic mutation): EXISTS
 - └─ intel-polymorphic-76 (76% of malware)
 - └─ intel-promptflux (LLM runtime rewriting)
 - └─ intel-vibeware (CSA "vibeware")
 - └─ intel-cleverhans-worm* (each instance differs)
- └─ ev-3 (AI real-time evasion): PARTIAL
 - └─ intel-promptsy (anti-uninstall overlays)
 - └─ intel-canfail (decoy code)

SDLC / WEAPONS PIPELINE

- └─ sd-3 (Auto field-to-foundry loop): GAP
 - └─ (the critical integration gap – see Portfolio Strategy)

└─ * = CleverHans update pending commit

3.4 Gen 2 Dependency Map

Gen 2 Capability	← Depends on Layer 1	Status
G2-1.1 Multi-implant mgmt	← ao-2, c2-1, c2-4, ao-5	PARTIAL
G2-1.2 Risk-tiered deployment	← ao-1, ao-3*, sd-3*	RESEARCH
G2-2.1 Sacrificial recon	← ao-2, c2-1/2/3, ev-1	NEAR-EXISTS
G2-2.2 EDR-specific tuning	← ev-1, ev-3, G2-2.1	PARTIAL
G2-3.1 Autonomous reporting	← ao-2, sd-3*	PARTIAL
G2-3.2 Custom exploit gen	← sd-3*	RESEARCH
G2-4.1 Strategic retreat	← c2-1, pc-1, ao-3*	PARTIAL
G2-4.2 Dormancy scheduling	← ao-5, c2-5*	PARTIAL

* = Layer 1 GAP blocking Gen 2 maturity

4. Gap Scorecard — Current State

Layer 1 (after CleverHans, pending commit)

Status	Count	Change from Infosec Europe
EXISTS	17	Unchanged
PARTIAL	14	Was 12 → +2 (ao-1, ao-5 from CleverHans)
GAP	3	Was 5 → -2

Remaining RED gaps:

ID	Capability	Blocking	Timeline estimate
ao-3	Environment value assessment	G2-1.2, G2-4.1	12-24 months (research)
c2-5	Legitimate user traffic mimicry	G2-4.2	12-18 months (ML infrastructure)
sd-3	Auto field-to-foundry loop	G2-1.2, G2-3.1, G2-3.2	6-12 months (integration)

Plus two capabilities that remain GAP but are sub-items of orchestration: - ao-4 (CTF objective matching) - ip-1 (Selective IP targeting)

Gen 2 (new, not yet in tracker)

Status	Count
NEAR-EXISTS	1 (G2-2.1 sacrificial recon)
PARTIAL	4 (G2-1.1, G2-2.2, G2-3.1, G2-4.1)
RESEARCH	3 (G2-1.2, G2-3.2, G2-4.2)

5. Pending Actions

Immediate — commit to GitHub tracker

Action	Source Document	What to update
Add intel-cleverhans-worm card	CleverHans_Research_Update_Gap_Changes.md § 4	data/intel.json
Move ao-1 GAP→PARTIAL	CleverHans_Research_Update_Gap_Changes.md § 5.1	data/gaps.json
Move ao-5 GAP→PARTIAL	CleverHans_Research_Update_Gap_Changes.md § 5.2	data/gaps.json
Add changelog entry	CleverHans_Research_Update_Gap_Changes.md § 7	data/changelog.json
Update scorecard in README	—	README.md

Short-term — extend the tracker (Gen 1 only)

Action	Source Document	Complexity
Add Gen 2 capabilities to tracker	BLOCKED — publish Gen 2 specification first	—
Add portfolio strategy view	BLOCKED — publish Gen 2 specification first	—
Update Framework Skill with Gen 2 methodology	BLOCKED — publish Gen 2 specification first	—
Update Research Skill with Gen 2 search strategies	BLOCKED — publish Gen 2 specification first	—
Write CleverHans LinkedIn post	—	Low — "the thesis is no longer theoretical"

Current priority — Gen 2 capability specification (separate workstream)

Action	Source	Output
Define Gen 2 capability specification	Gen2_Capabilities_Framework.md , Hydra_Portfolio_Strategy.md	Whitepaper: what a Gen 2 weapon requires
Nuclear proliferation framing paper	All evolved docs	Publication: Gen 1 = components, Gen 2 = integration, Gen 3 = deployment
Do NOT tag current intel against Gen 2	—	Gen 2 intel tagging happens AFTER publication
Do NOT modify Gen 1 capability list	—	34 capabilities are fixed

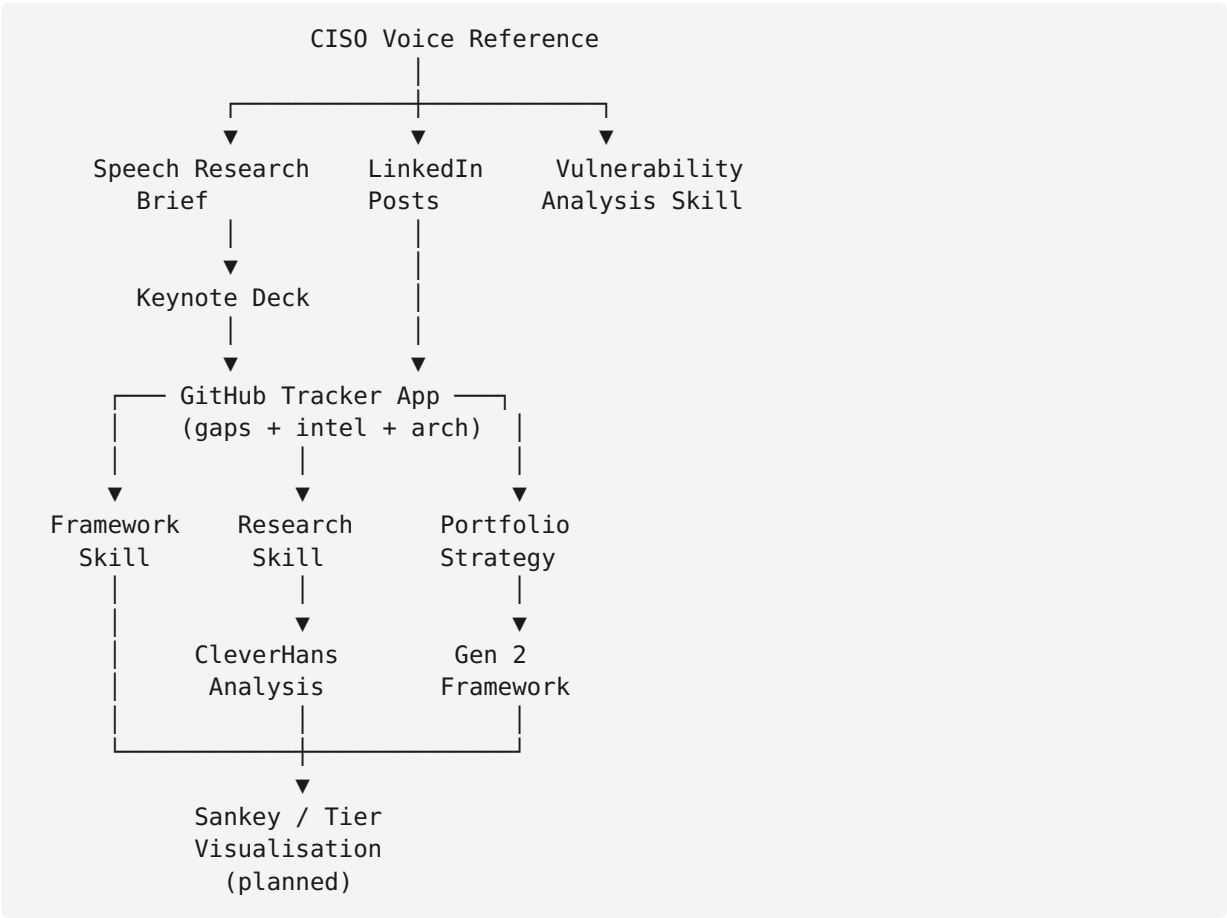
Medium-term — evolve the framework

Action	Source	Complexity
Build Sankey visualisation (probability flows)	Gen2_Capabilities_Framework.md	High — interactive, time-series
Resolve ao-1 naming conflict (attack surface vs recursive reasoning)	CleverHans analysis	Low — clarify framework definitions
Add tier-differentiated timeline to tracker	Gen2_Capabilities_Framework.md	Medium — Tier 1-5 capability maturity curves
Consider splitting ao-5 into two sub-capabilities (field intel vs evasion)	CleverHans_vs_Hydra_Gap_Analysis.md	Low — already identified as composite

Planned Publications

Publication	Framing	Status
CleverHans LinkedIn post	"The thesis is no longer theoretical" — peer-reviewed POC validates orchestration layer	Pending
Gen 2 Paper: Nuclear Proliferation Framework	Full paper framing AI offensive capability tracking as nuclear proliferation monitoring. Gen 1 = component proliferation. Gen 2 = system integration testing. Gen 3 = operational deployment. Dual-use problem. Breakout timelines by tier. DEFCON readiness model	Planned
Portfolio Strategy paper	Chess model — multi-stage, multi-campaign, Pawns/Knights/King. Why single-campaign models underestimate the threat	Planned — may combine with Gen 2 paper
CVE Noise / Vulnerability Prioritisation article	36→4 reduction, decision tree, CVSS vs reality. Standalone LinkedIn article	Planned

6. Artifact Dependency Graph



7. Cross-Reference Index

By topic — where to find information on:

Topic	Primary artifact	Supporting artifacts
The convergence thesis	Speech Research Brief § Opening, Act 1	Keynote slides 5–7
Three lanes (worms/ ransomware/IP theft)	Speech Research Brief § Act 1	Keynote slide 6
Double extortion reframe	Speech Research Brief § Lane 2	Keynote slide 6 (updated)
TTP crossover (70–80% overlap)	Speech Research Brief § TTP Crossover	—
The architecture (4 layers)	Speech Research Brief § Act 3	Keynote slides 13–14, Tracker Architecture tab
The weapons foundry / SDLC	Speech Research Brief § Act 3, Portfolio Strategy § Stage 2	Tracker Architecture tab
Portfolio strategy (chess model)	Hydra Portfolio Strategy	—
Gap analysis (Layer 1)	Framework Skill § 7.4, Tracker Gap Analysis tab	CleverHans Research Update
Gen 2 capabilities	Gen2 Capabilities Framework	CleverHans Gap Analysis § Part 3
Sankey / probability model	Gen2 Capabilities Framework § Sankey	—
CleverHans findings	CleverHans Research Update	CleverHans Gap Analysis
Mini Shai-Hulud case study	Speech Research Brief § Case Study	Keynote slide 13
PROMPTSPY / GTIG findings	Tracker intel cards, Framework Skill § 7.4	—
CVE noise / prioritisation	Why_Raw_CVE_Counts_Are_Noise.docx	LinkedIn post (published)
Patching / CE / pen test stats	Speech Research Brief § Opening	Keynote slides 2–4
Tier mapping (who can build this)	Speech Research Brief § Act 4, Portfolio Strategy § Tier implications	Keynote slide 16
How to update the tracker	Framework Skill §§ 2–6, Research Skill §§ 2–4	README.md in GitHub repo
How to research new intel	Research Skill §§ 2–5	—
How to write intel cards	Framework Skill § Step 4	—
How to write LinkedIn posts	CISO Voice Reference §§ 3–5	—

8. Version History

Version	Date	Changes
1.1	2026-06-06	Gen 1/2/3 doctrine formalised. Nuclear proliferation framing added. Evidence thresholds defined (offensive only — enterprise IT is context, not evidence). Planned publications section added including Gen 2 paper.
1.0	2026-06-06	Initial roadmap. 25 intel entries, 34 Layer 1 capabilities, 8 Gen 2 capabilities. CleverHans analysis complete. Portfolio strategy and Gen 2 framework documented.

This document is the project's table of contents. When starting a new session, read this first to understand what exists and where to find it.